Hazard Analysis and ASIL Determination for Airbag ECU

1. Introduction to Functional Safety and Item Definition

In accordance with ISO 26262-3, "Functional Safety" is defined as the absence of unreasonable risk due to hazards caused by the malfunctioning behavior of electrical and electronic (E/E) systems. This compliance report establishes the safety mandate for the Airbag Electronic Control Unit (ECU), a high-integrity system where random hardware failures or systematic faults can lead to life-critical consequences.The "Item" under analysis is the Airbag ECU. Its primary function is the execution of a safety-critical mission: continuous crash detection via a redundant sensing path comprising peripheral impact/pressure sensors and internal g-sensors and the execution of command logic for the deployment of airbags and seatbelt pretensioners. The objective of this report is to define the Item's boundaries, perform a Hazard Analysis and Risk Assessment (HARA), and establish ASIL-rated Safety Goals (SGs) that dictate the subsequent Technical Safety Concept (TSC).

2. System Architectural Overview

A granular evaluation of the ECU's technical architecture is mandatory to identify the malfunction pathways that compromise safety requirements. In an ASIL D environment, we must account for every potential failure point within the hardware and communication blocks.The Airbag ECU technical architecture integrates the following elements:

- **Processing Units:** A high-performance microcontroller featuring a dedicated CPU, an external Watchdog for execution monitoring, and a Backup Power circuit (capacitive energy storage) to ensure deployment capability even after battery disconnection during an impact.
- **Communication Interfaces:** The system relies on the Serial Peripheral Interface (SPI) for synchronous communication between the microcontroller (acting as the SPI Master) and its peripheral sensing/actuation ICs, alongside CAN bus integration for vehicle-level diagnostics.
- **Sensing Array:** A multi-domain array utilizing redundant sensing paths, including upfront impact sensors, side-impact pressure sensors, and internal tri-axial acceleration sensors.
- **Actuation Path:** High-side and low-side power stages that control the firing loops for the deployment of airbags and pretensioners.The Serial Peripheral Interface (SPI) is a critical focus of this analysis. As the SPI Master, the ECU's internal shifter logic and baud rate generator are vital; any corruption here such as a "fire" command issued due

to corrupted noise or a wrong decode of clock selection can lead directly to an unintended deployment. This architectural complexity necessitates a structured approach to Hazard Identification.

3. Hazard Identification and Malfunction Analysis

The HARA process, mandated by ISO 26262-3, drives the identification of potential system malfunctions. By mapping technical failure modes to operational risks, we define the boundaries of safe operation. Based on the IJERT FMEDA and Fraunhofer system-level modeling, the following primary hazards have been identified:

1. **Unintended airbag deployment:** A malfunction resulting in activation without a valid crash event. This is often linked to **shifter logic corruption** or **data corruption in the transmit register**, where a malformed SPI message is interpreted as a firing command. Worst-case effect: Loss of vehicle control and direct high-velocity trauma.
2. **Failure to deploy during a crash:** A malfunction where the system remains inert during a valid collision. Technical causes include failures in the **baud rate generator** or **protocol generation logic** (SCLK/SS failures) preventing sensor data acquisition. Effect: Total loss of passive safety, resulting in life-threatening or fatal injuries.
3. **Passenger airbag deployment with a baby seat present:** A failure to **suppress** deployment when the sensing array indicates a rear-facing infant seat. This represents a failure in the logic-gate processing of the occupancy sensor data. Effect: High risk of infant fatality.
4. **Delayed deployment:** Activation occurring outside the Fault Tolerant Time Interval (FTTI). If deployment exceeds the timing requirements (e.g., due to protocol delays or interrupt service failures), the occupant may already be out of position, making the airbag a source of injury rather than protection.
5. **Incorrect airbag selection during activation:** The system deploys an inappropriate squib (e.g., right-side curtain for a left-side impact). This is typically caused by **wrong decode of clock selection** or **interrupt count errors**, leading to improper actuation logic.

4. Risk Assessment Methodology: Severity, Exposure, and Controllability

The ISO 26262 framework classifies risk by evaluating the potential harm through the lens of environmental exposure and human-led mitigation.

- **Severity (S0-S3):** Quantifies the intensity of potential harm. For the Airbag ECU, failures are typically S3, indicating life-threatening or fatal injuries.
- **Exposure (E0-E4):** Measures the probability of the vehicle being in a situation where the hazard can occur. E4 represents "highly probable" scenarios (e.g., high-speed driving).
- **Controllability (C0-C3):** Assesses the driver's ability to prevent injury once a malfunction occurs. Unintended airbag deployment is classified as C3 (uncontrollable), as a driver cannot mitigate a high-velocity deployment while in motion.**Strategic Justification:** The combination of S3, E4, and C3 variables mandates an **ASIL D** classification. This rating represents the highest rigor in automotive safety assurance,

requiring the implementation of advanced hardware metrics and rigorous diagnostic coverage as defined in ISO 26262-5.

5. ASIL Determination and Classification Matrix

The following table summarizes the ASIL assignments for the Airbag ECU. Assignments are driven by the catastrophic nature of passive safety failures.

Hazard	Severity(S)	Exposure(E)	Controllability(C)	ASIL Level
Unintendted Deployment	S3	E4	C3	D
No Deployment during Crash	S3	E4	C3	D
Suppression Failure(Baby Seat)	S3	E3	C3	C
Delayed Deployment	S3	E4	C2	C
Wrong Airbag Activation	S3	E4	C3	D

Airbag ECU ASIL Classification

Note: While SG3 and SG4 are rated ASIL C due to lower exposure (frequency of baby seats) or higher controllability (timing delays), the overall ECU design is governed by the ASIL D mandate of the primary firing logic.

6. Functional Safety Concept: Safety Goals (SGs)

Safety Goals represent top-level requirements derived as the inversion of identified hazards. These goals establish the mandate for the system's "Safe State"—which for the Airbag ECU is typically "deployment inhibited" or "firing loop isolation."

- **SG1:** Prevent unintended deployment — **ASIL D**
- **SG2:** Ensure deployment occurs within the FTTI during a valid crash — **ASIL D**
- **SG3:** Suppress passenger airbag deployment when a baby seat is detected — **ASIL C**
- **SG4:** Ensure correct airbag selection based on impact zone — **ASIL D**SG3 is a vital logic requirement: the safety concept must ensure that a "Suppress" command overrules "Fire" logic when specific occupancy criteria are met. Any failure in this suppression logic is a direct violation of the Functional Safety Concept.

7. References

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